ECON-7102: Environmental Economics I

Class 15: Value of a statistical life and Lee & Taylor (2019)

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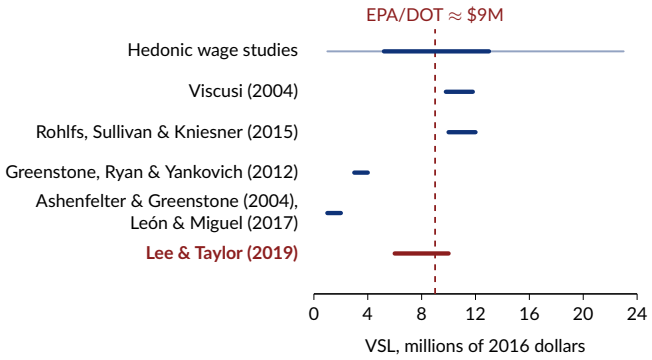
Fall 2026

Today's agenda

1. Questions?
2. Housekeeping:
   - Midterm: Wed Oct 21, in class.
3. Today:
   - Where the VSL came from (Banzhaf, JEP 2014)
   - Valuing mortality risk (P&R Chapter 20)
   - Lee and Taylor (AEJ: Policy, 2019)
4. After the midterm...

What is a life worth?

- Mortality is up to 70% of monetized benefits of major federal rules.
- Agencies use ≈ $9M per life.
- Estimates span $1M to $23M+.
- **Q:** What is a “statistical” life?



Origins of the VSL (Banzhaf 2014)

1950: RAND

‘‘*...it sought to solve a classic problem formulated in terms of choosing bombs and bombers to maximize damage, subject to a fixed dollar budget (to procure, operate, and maintain the force) and fixed budget of fissile material...*”
Bombing study puts zero weight on aircrew lives. The Air Force was not very happy...

1963: Carlson

Values pilots’ lives from Air Force choices. Treats hazard pay as risk.

1968: Schelling

Value small risk reductions to the people who bear them. Sum to “statistical” lives.
“...but people have been dying for as long as they have been living; and where life and death are concerned we are all consumers. We nearly all want our lives extended and are probably willing to pay for it.”

> *“Worth to whom? … I shall propose that it is to the people who may die.”*
> — Schelling (1968)

Definition (P&R 20.2)

- 10,000 people each pay $500 to reduce their risk by 1/10,000.
- Total $5M. One fewer expected death.

$$\text{VSL} = \frac{\$500}{1/10{,}000} = \$5 \text{ million}$$

Definition 20.1. The VSL is the amount a population would pay ex ante to prevent the death of a single unknown member, during a given time period.

Static model (P&R 20.1.1)

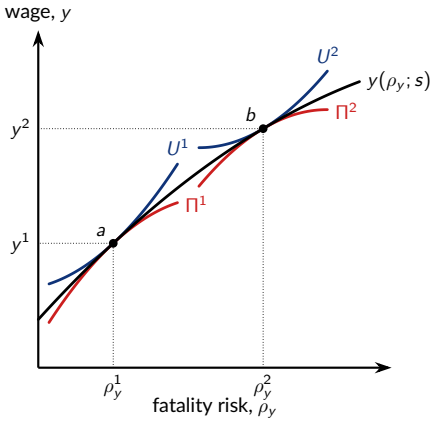
- Utility $U(z; s)$ if alive, 0 if dead. Three small risks:
  - ρ_e: exogenous (e.g., a contaminant)
  - $\rho(x)$: safety good x at price p, with $\rho'(x) < 0$
  - ρ_y: job risk, paid wage $y(\rho_y; s)$ with $y'(\rho_y) > 0$
- Survival $\pi \approx 1 - \rho_e - \rho(x) - \rho_y$, budget $y = px + z$:

$$\max_{x,\, \rho_y} \ \bar{U} = \pi\, U\big(y(\rho_y; s) - px; s\big)$$

- FOCs $\pi U_z \frac{dy}{d\rho_y} = U$ and $-\rho'(x)\, U = \pi U_z\, p$ give:

$$\underbrace{\frac{dy}{d\rho_y}}_{\text{extra pay for riskier job}} = \underbrace{\frac{U}{\pi\, U_z}}_{\text{utility of life in dollars}} = \underbrace{-\frac{p}{\rho'(x)}}_{\text{cost of buying lower risk}}$$

$\Longrightarrow$ **Wage gradient = MWTP for risk reduction.**



Life cycle model: age and latency (P&R 20.1.2)

- MWTP today for lower risk at age k:

$$MWTP_{ak} = \pi_{ak}(1+r)^{a-k}\, MWTP_{kk} \qquad (20.26)$$

  where:
  - a: current age, k: age at which the risk change occurs ($k > a$)
  - π_{ak}: probability of surviving from age a to age k
  - r: risk-free interest rate for borrowing and saving
  - $MWTP_{kk}$: MWTP at age k for a risk change at age k
- Latency: discount by survival to k and by r.
- Age: MWTP can increase or decrease with age.
- **Q:** Environmental deaths are mostly older adults and illness. Does a VSL from 35-year-old factory workers transfer to a 75-year-old grandmother? To an infant?

Hedonic wage models (P&R 20.2.1)

- $\ln y_i = \beta X_i + \gamma \rho_i^y + \varepsilon_i, \quad \text{VSL} = \hat{\gamma} \times \bar{y} \times 2{,}000 \times 100{,}000$

- **Problem 1:** risk is an industry average, not the worker's plant (measurement error).
- **Problem 2:** risky jobs differ in many other ways (omitted variables).
- Problem 3: do workers know the risks?
- Problem 4: does a VSL from prime-age blue-collar men transfer?

$\Longrightarrow$ Problems 1 and 2 motivate Lee and Taylor.

Lee & Taylor (2019): setting and data

- **OSHA inspections**: surprise inspections focusing on safety. Violations must be fixed.
- **1987–1997**: randomly assigned within industry $\times$ state $\times$ year.
- **Data**: OSHA records + confidential Census of Manufactures.
  - 65,300 plants, 1987–1997
  - Plant-level wages, fringe benefits, and fatalities

Event studies

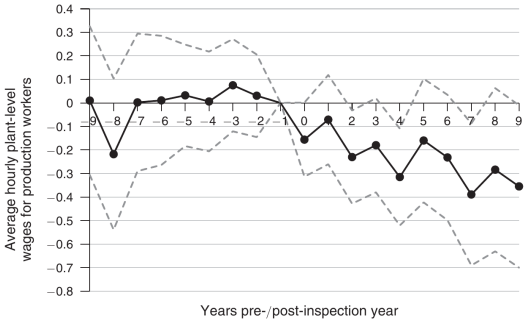


Figure 2. Event Study Analysis of Compensating Wages

- No pre-trend. Wages decrease by $0.20–0.30/hr.

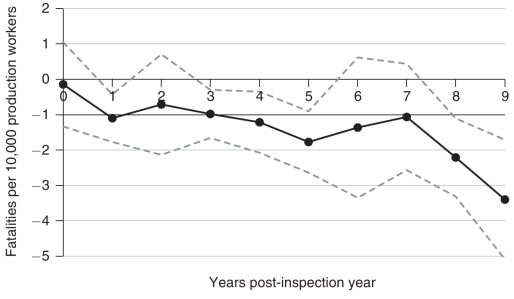


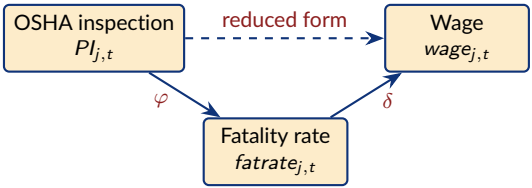
Figure 3. Post-inspection Event Study Analysis of Fatality Rates

- Fatalities decrease by 1–2 per 10,000 workers.

Empirical strategy: IV

- First stage: $fatrate_{j,t} = \varphi PI_{j,t} + P_j + I_{j,t} \times S_s \times T_t + \epsilon_{j,t}$
- Second stage: $wage_{j,t} = \delta \widehat{fatrate}_{j,t} + P_j + I_{j,t} \times S_s \times T_t + u_{j,t}$
  - P_j: plant fixed effects. Absorb permanent differences across plants (e.g., baseline safety, location, workforce).
  - $I_{j,t} \times S_s \times T_t$: industry-by-state-by-year fixed effects. Compare inspected and uninspected plants in the same industry, state, and year, which is the level at which OSHA randomized.
- VSL $= \hat{\delta} \times 2{,}000 \times 10{,}000$

- **Q:** What else could an inspection change that affects wages?



Results: first and second stage

- Inspection: -1.4 deaths per 10,000 ($\approx -45\%$).
- Wages: \$0.20–0.24/hr per death per 10,000.
- Fringe adds $\approx$ 40%.

TABLE 2—IV ESTIMATES OF PLANT-LEVEL CHANGES IN RISKS AND WAGES IN RESPONSE TO RECEIVING AN OSHA INSPECTION

Panel A. Select coefficient estimates for first-stage regression (equation (3))[a]					
Programmed inspections ($PI = 1$ in year of inspection and each year thereafter; 0 otherwise)	−1.338	−1.339	−1.398	−1.394	−1.270
	(0.338)	(0.338)	(0.309)	(0.309)	(0.314)
Model variations					
Plant characteristics included[b]	Yes	No	No	No	No
Weighting[c]	None	None	ASM × PW	ASM × PW	ASM × PW
R^2	0.296	0.296	0.294	0.294	0.294
F-statistic[d]	15.70	15.71	20.34	20.32	16.32
Panel B. Select coefficient estimates for second-stage regression (equation (4))[e]					
Fatality rate ($\widehat{fatrate}_{j,t}$)	0.204	0.223	0.242	0.0180	0.337
	(0.0715)	(0.0751)	(0.0795)	(0.00599)	(0.115)
Model variations					
Dependent variable	*Wages*	*Wages*	*Wages*	*ln(Wages)*	*Wages + Fringe*
Plant characteristics included[b]	Yes	No	No	No	No
Weighting[c]	None	None	ASM × PW	ASM × PW	ASM × PW
R^2	0.484	0.433	0.547	0.582	0.347

Results: VSL

- Wages: **$6–7M**
- With fringe: **$8–10M**
- Stable across samples. Wide CIs.
- **Q:** Whose MWTP is this?

TABLE 3—VSL ESTIMATES BY ESTIMATION SAMPLE AND MODEL
(REPORTED IN MILLIONS, 2016 DOLLARS)

VSL estimates (95 percent confidence intervals)					
Panel A. Baseline sample used in estimation (year of inspection and following year excluded)					
Observations = 252,800 and Number of Plants = 65,300	$5.69	$6.21	$6.74	$6.38	$9.39
	(1.78–9.59)	(2.11–10.32)	(2.40–11.09)	(2.22–10.54)	(3.09–15.69)
Panel B. All years (1987–1997) included in the estimation sample					
Observations = 257,600 and Number of Plants = 65,700	$6.02	$6.52	$6.49	$5.99	$10.00
	(1.53–10.51)	(1.85–11.20)	(0.98–12.01)	(0.95–11.03)	(1.83–18.16)
Panel C. Year of inspection excluded from estimation sample					
Observations = 254,500 and Number of Plants = 65,300	$6.33	$6.86	$5.82	$5.32	$7.67
	(1.54–11.11)	(1.85–11.86)	(0.83–10.82)	(0.68–9.96)	(1.57–13.78)
Panel D. Baseline sample, excluding ASM observations					
Observations = 133,600 and Number of Plants = 63,300	$6.80	$7.27	$6.83	$6.91	$7.93
	(1.43–12.17)	(1.65–12.90)	(1.96–11.70)	(2.13–11.70)	(2.20–13.67)
Panel E. Estimation sample includes all years (1987–1997) and plants inspected prior to 1987					
Observations = 499,000 and Number of Plants = 110,000	$6.35	$6.66	$7.50	$7.16	$9.82
	(2.94–9.77)	(3.15–10.17)	(2.85–12.14)	(2.83–11.50)	(3.43–16.21)
Model variations					
Dependent variable	*Wages*	*Wages*	*Wages*	*ln(Wages)*	*Wages + Fringe*
Plant characteristics included	Yes	No	No	No	No
Weighting method	None	None	ASM × PW	ASM × PW	ASM × PW

Conventional hedonic wage models, same data

- OLS VSL: $107M to $13K.
- Industry risk: confounded with inter-industry wage differences.
- Plant risk: noisy, attenuated.

TABLE 6—VSL ESTIMATES ARISING FROM ALTERNATIVE AND UN-INSTRUMENTED RISK RATES (REPORTED IN MILLIONS, 2016 DOLLARS)

	VSL estimate (95% confidence intervals)		
Fatality rates measured as			
Annual average for the industry at the two-digit SIC level	$107.4 (99.5–115.3)	$10.82 (3.61–18.03)	$1.48 (−4.26–7.22)
Annual average for the industry at the four-digit SIC level	$13.80 (12.10–15.50)	$5.16 (3.74–6.58)	$0.050 (−0.80–0.90)
Annual average for each plant	$0.046 (0.015–0.077)	$0.030 (0.005–0.055)	$0.013 (−0.013–0.039)
Model variations			
Industrial sector fixed effects (two-digit SIC level)	No	Yes	No
Plant fixed effects	No	No	Yes
Plant characteristics included[a]	Yes	Yes	Yes

Robustness and limits

- No spillovers to nearby or sister plants.
- Placebo: non-production wages do not change.
- Inspections are costly: TFP decreases by $\approx 1.6\%$.
- Unobserved injury risk could inflate the VSL.
- Sample: manufacturing production workers, accidental deaths.

Takeaways

- VSL = summed MWTP for small risk changes $\neq$ the “value of a life”.
- Hedonic wage slope = MWTP for risk reduction.
- Cross-sectional estimates are fragile.
- Randomized inspections: VSL of \$6–10M.
- Transfer to environmental deaths still needs theory.

For next class...

- **Wed Oct 21 (Class 16):** Midterm, in class.
- **Mon Oct 26 (Class 17–onwards):**
  - Reading papers and discussing them!
  - I am going to tweak the readings/schedules.
  - Students will *lead discussion* (not lecture!) for one paper each class (vote on preferences forthcoming).